Question 22 (3 marks)

An animal classification tool is to be developed. The classes to be used are shown below.

Mammal
Colour
Size
Birth type
Number of teeth

Reptile
Colour
Size
Scale type
Tongue type

Whale
Colour
Size
Birth type
Number of teeth
Blowhole type
Migration location

Dog
Colour
Size
Birth type
Number of teeth
Breed
Fur

Construct a class diagram that shows inheritance and removes redundant attributes for each class. Additional classes may be included if required.

You do NOT need to show the nature of the relationships between the classes.

Full screen Reset

Library

To move canvas, hold mouse wheel or spacebar while dragging, or use the hand tool

100%

Question 21 (4 marks)

Below is the *Issues* table for a technical support company. FixTime, in minutes, is the time taken to complete each job.

JobID	Issue	Response	FixTime	Category	Status
0001	Cannot connect to the internet	Clear cache	22	Software	Pending
0002	Installation failure	Reinstall the printer drivers	76	Hardware	Pending
0004	Forgot password	Reset password	97	Software	Resolved
0005	Installation failure	Follow the installation guide	110	Hardware	Escalated
0009	Blue screen error	Run a system diagnostic test	120	Software	Pending
0011	Installation failure	Verify email settings	45	Software	Escalated

Construct a SQL query that displays:

Full screen Run Reset DB

- **the number of pending software jobs**
- **the number of pending hardware jobs**
- **the number of escalated software jobs**
- **the number of escalated hardware jobs**

List the results in order from the highest to lowest, based on the number of jobs.

Query

```
1
```

No results to display

Question 20 (3 marks)

Describe how session management can affect the efficiency of code execution for the user.

0 / 45 words

Previous question

Next question

Question 19 (3 marks)

In an organisation, temporary passwords are automatically generated for new user accounts. Each password is created from the first three letters of the username followed by '123'.

For example, the username 'NESA' will have the password 'NES123'.

When a user logs in for the first time, they are asked for their username and password. They receive a welcome message or an error message, depending on whether the password entered follows the rules of new user accounts.

Write a Python program that asks a new user for their username and password, and then checks whether the password rule has been followed.

Full screen Run

```
1 # Input code
2
```

```
# Output
```

Question 18 (3 marks)

A streaming service has exclusive rights to broadcast a live event each Sunday afternoon.

Explain how the streaming service can use metadata and data mining to prevent technical difficulties during the broadcast of this weekly event.

0 / 60 words

‹ Previous question

Next question ›

Question 17 (3 marks)

An algorithm is required by a school to rate the risk of a location for school excursions.

A teacher completes a 10-question survey to assess potential hazards for the location. Each question is given a score from 1 (low) to 5 (high), and the scores are stored in a 10-element integer array called *Risks*.

The scores for each question are added to give a total risk score.

The excursion risk is rated as:

- Low (total less than 20)
- Medium (total between 20 – 35)
- High (total above 35).

Using pseudocode, with a repeat-until loop, write an algorithm to determine the excursion risk rating for a location. You can assume that the scores for the 10 questions are already stored in the array *Risks*.

Full screen

```
1
```

Previous question

Next question

Question 16 (9 marks)

A weather app is to be designed to display real-time updates such as:

- rain commencing within the next 10 minutes
- weather event warnings and warning cancellations
- location specific weather conditions.

(a) Compare the use of a progressive web app (PWA) with an interactive website, for delivering this weather app. (3 marks) hide

0 / 65 words

(b) Outline TWO ways, other than the use of a PWA, to minimise the load time of this weather app. (3 marks) hide

0 / 35 words

(c) Describe ONE way that the 'privacy by design' approach could be implemented in this weather app. (3 marks) hide

0 / 50 words

Question 15 (6 marks)

hide

A computer game is being developed.

(a) Part of a data dictionary for the computer game is shown.

Select the correct entries to complete the data dictionary. (3 marks)

Variable	Data type	Format for display	Size in bytes	Description	Example
PlayerId		NXXXXN		Identifies player	
PlayerName		XX...XX	15	Player name	Nat Kim
Level		NN	1	Level of player	27
OnMission		X	1 bit	Y or N	Y
LastSave	Date and Time		12	Time last saved 24-hour time	13:45:21.221

hide

(b) While this game is being developed, there are persistent issues with the code.

Outline TWO software debugging tools that can be used to identify these logic errors. (3 marks)

0 / 60 words

Question 14 (1 mark)

Consider this code fragment.

```
FOR index = 5 TO 3
    word[index] = word[index-2]
NEXT index
```

The array, word, contains the following data.

index	1	2	3	4	5
word[index]	S	M	A	R	T

Select the correct item from each dropdown menu to show the contents of the array after the code is run.

Previous question Next question

Question 13 (2 marks)

A software developer plans to use static application security testing (SAST) and dynamic application security testing (DAST) to check an application that is being developed.

For each strategy, choose the most appropriate stage in software development for it to be applied. Indicate whether access to source code is required.

Strategy	Timing in software development	Access to source code
SAST		
DAST		

Previous question

Next question

Question 12 (2 marks)

For which of the following situations would logistic regression be used when training a machine learning model to make predictions?

Select all that apply.

- [ ] Identifying whether a received email is spam
- [ ] Determining the outcome of a two-party election
- [ ] Calculating house prices based on size of the land
- [ ] Investigating the relationship between height and weight
- [ ] Estimating the rate at which a contagious illness will spread
- [ ] Working out if someone will be admitted to a university course

Previous question

Next question

Question 11 (1 mark)

Which of the following are key principles of DevOps?

Select all that apply.

- [ ] Rapid feedback
- [ ] Freedom from bias
- [ ] Procedural fairness
- [ ] Automation of repetitive tasks
- [ ] Collaboration and shared ownership

‹ Previous question

Next question ›

Question 10 (1 mark)

Which of the following are goals of regulatory compliance in relation to software development?

Select all that apply.

- [ ] Limiting the need for software updates
- [ ] Protecting sensitive data from unauthorised access
- [ ] Allowing developers to bypass security features when needed
- [ ] Reducing the testing time required during software development
- [ ] Ensuring software applications meet legal and industry standards

Previous question

Next question

Question 9 (1 mark)

Two approaches that can be used when programming databases for use in web applications are Object-Relational Mapping (ORM) and Structured Query Language (SQL).

Match each characteristic to the appropriate approach.

	Object-Relational Mapping (ORM)	Structured Query Language (SQL)
Code is focused on object manipulation	○	○
Tables can be joined to enhance searching	○	○
Programmers can choose the language used to access the database	○	○

‹ Previous question

Next question ›

Question 8 (1 mark)

File locking prevents multiple users from editing a file at the same time. When a shared file is opened by a user, file locking ensures that only this user can make changes to the file while it is locked.

Which of the following does file locking minimise?

- ○ Race conditions
- ○ Invalid redirection
- ○ Cross-site scripting
- ○ Authentication issues

‹ Previous question

Next question ›

Question 7 (1 mark)

Consider the following cascading style sheet (CSS) code fragment for a website.

```
body {
    font-size: 100%;
    font-style: italic;
    line-height: 1.8;
    font-weight: normal;
}
h1 {
    text-align: center;
    font-style: normal;
    font-size: 250%;
    letter-spacing: 5px;
    background-color: rgb(0, 0, 100);
    color: white;
    font-weight: 900;
    border: 3px solid green;
}
```

Which of the following best describes the appearance of the heading on the website?

- ○ White background, black text, green border
- ○ Centre aligned, italic, increased letter spacing
- ○ Blue background, white text, border around the heading
- ○ Left aligned, bold, and 2.5 times larger than the rest of the page

‹ Previous question Next question ›

Question 6 (1 mark)

Which of the following is a function of TLS?

- ○ Delivering packets between hosts on the internet
- ○ Dropping malformed packets before they reach a host computer
- ○ Adding headers to data packets containing authentication information
- ○ Ensuring that parties exchanging information are who they claim to be

‹ Previous question

Next question ›

Question 5 (1 mark)

What is a potential vulnerability of an application programming interface (API)?

- ○ Excess load times
- ○ Biases in machine learning
- ○ Exposure of more data than intended
- ○ Inappropriate use of screen elements

‹ Previous question

Next question ›

Question 4 (1 mark)

Fragment 1 and Fragment 2 are Python code that achieve the same purpose.

Fragment 1

```
numbers = [1, 2, 3, 4, 5]
total = 0
for num in numbers:
    total += num
print(total)
```

Fragment 2

```
numbers = [1, 2, 3, 4, 5]
print(sum(numbers))
```

Fragment 2 is an example of

- recursive code.
- list comprehensions.
- language-dependent code optimisation.
- using a hash table to create shorter solutions.

Previous question Next question

Question 3 (1 mark)

An image recognition program uses machine learning (ML) to recognise letters. It is trained using a range of images of handwritten letters. Each of the images below is labelled with the letter B when training the program.



What model of ML training is used to train this software?

- ○ Supervised learning
- ○ Unsupervised learning
- ○ Reinforcement learning
- ○ Semi-supervised learning

‹ Previous question

Next question ›

Question 2 (1 mark)

Complete the statement by selecting the correct term from each dropdown menu.

The [ ] approach typically requires extensive documentation, while the [ ] approach often emphasises working software over documentation.

Previous question

Next question

Question 1 (1 mark)

A streaming platform updated its content suggestion algorithm for users. A group of users was upgraded to the new algorithm, and feedback was collected before replacing the algorithm for all users.

Which implementation method was used?

- (●) Pilot
- ( ) Direct
- ( ) Phased
- ( ) Parallel

‹ Previous question

Next question ›

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2025 HSC Software Engineering

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‹ Previous question
Next question ›

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hide

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LastSave	Date and Time		12	Time last saved 24-hour time	13:45:21.221

hide

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0 / 60 words

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0 / 65 words

(b) Outline TWO ways, other than the use of a PWA, to minimise the load time of this weather app. (3 marks) hide

0 / 35 words

(c) Describe ONE way that the 'privacy by design' approach could be implemented in this weather app. (3 marks) hide

0 / 50 words

Previous question Next question

Question 23 (9 marks)

A food delivery service offers prepackaged meals tailored to different dietary preferences, for example, vegetarian and gluten-free. It has developed a website for customers to place orders. The website is ready for testing and feedback.

The website includes:

- a menu that dynamically updates based on meal availability
- user accounts where customers can track deliveries and manage subscriptions
- a backend that processes orders and payments, updates stock and communicates with the delivery team.

(a) Outline potential social and ethical issues associated with the use of this website. (3 marks) hide

0 / 50 words

(b) Explain why server-side web programming would be needed to develop this website.

Support your answer with examples from the website. (3 marks) hide

0 / 65 words

(c) Security features have been included in the design of the website. Choose the fundamental security concept that matches each of the security features. (3 marks) hide

Feature	Concept
The system verifies that the total price displayed matches the actual total after items are added to the cart.	
The backend ensures the website remains accessible during peak order times.	
The system logs all user actions, such as changes to subscriptions or delivery addresses.	
User payment details, such as credit card numbers, are encrypted during transactions.	
A user with a basic account can place orders but cannot access admin features.	

Question 27 (6 marks)

Watch the video, and then answer the question.

Due to copyright restrictions, this material cannot be displayed until permission has been obtained.

Explain how human bias could affect the usefulness of artificial intelligence (AI) solutions.

Support your answer with reference to the video.

0 / 185 words

Previous question

Exit exam

Question 26 (5 marks)

As part of a computer game, a program is needed to randomly place 15 diamonds on a 5 × 5 grid. Once the locations of the diamonds have been generated, the program outputs the 5 × 5 grid, with the location of each diamond signified by an 'X'.

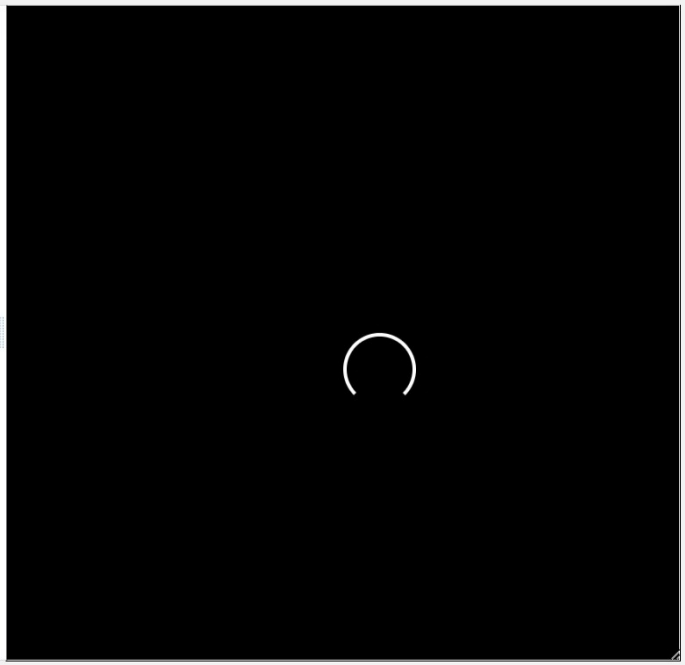
An example of the output required is shown below.

```
| X |   | X |   | X |
|   | X |   |   | X |
|   | X | X | X |   |
| X |   | X |   |   |
| X | X | X | X | X |
```

Using Python, write a program that achieves this task.

Full screen Run

```
1  # Input code
2
```



Question 25 (5 marks)

An airline introduces a virtual personal assistant as an initial support contact point for cancelled flights, missed connections, or lost luggage. The virtual personal assistant uses a neural network to:

- analyse support queries to direct customers to the appropriate department
- recommend appropriate tone and wording to be used by the human customer support representative
- assess data such as support turnaround time, customer ratings, and repeat contacts.

Explain how the neural network supports the virtual personal assistant. Include ONE positive and ONE negative impact of using the virtual personal assistant.

0 / 135 words

Previous question Next question

Question 24 (5 marks)

A construction company introduces a safety monitoring system that uses artificial intelligence. Cameras and wearable sensors track worker movements and environmental conditions. The system detects unsafe behaviours through artificial intelligence, such as:

- standing too close to operating machinery
- entering restricted zones
- failure to wear personal protective equipment (PPE).

Alerts are sent in real-time to site supervisors, and long-term data is stored for further analysis.

Discuss the possible impact of this monitoring system on workers.

0 / 110 words

Previous question

Next question

Question 23 (9 marks)

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0 / 50 words

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0 / 65 words

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